

```
In [1]: import seaborn as sns
```

```
In [2]: import warnings
warnings.filterwarnings('ignore', category=FutureWarning)
sns.get_dataset_names()
```

```
Out[2]: ['anagrams',
        'anscombe',
        'attention',
        'brain_networks',
        'car_crashes',
        'diamonds',
        'dots',
        'dowjones',
        'exercise',
        'flights',
        'fmri',
        'geyser',
        'glue',
        'healthexp',
        'iris',
        'mpg',
        'penguins',
        'planets',
        'seaice',
        'taxis',
        'tips',
        'titanic']
```

```
In [3]: tips = sns.load_dataset("tips")
tips.head()
```

```
Out[3]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
In [4]: titanic = sns.load_dataset("titanic")
titanic.head()
```

Out[4]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T



In [5]: tips

Out[5]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

In [6]: `sns.set_theme(style="darkgrid")`

In [8]: `tips.to_csv("tips_dataset.csv", index=False)`

In [10]: `import pandas as pd`
`import os`
`os.getcwd()`

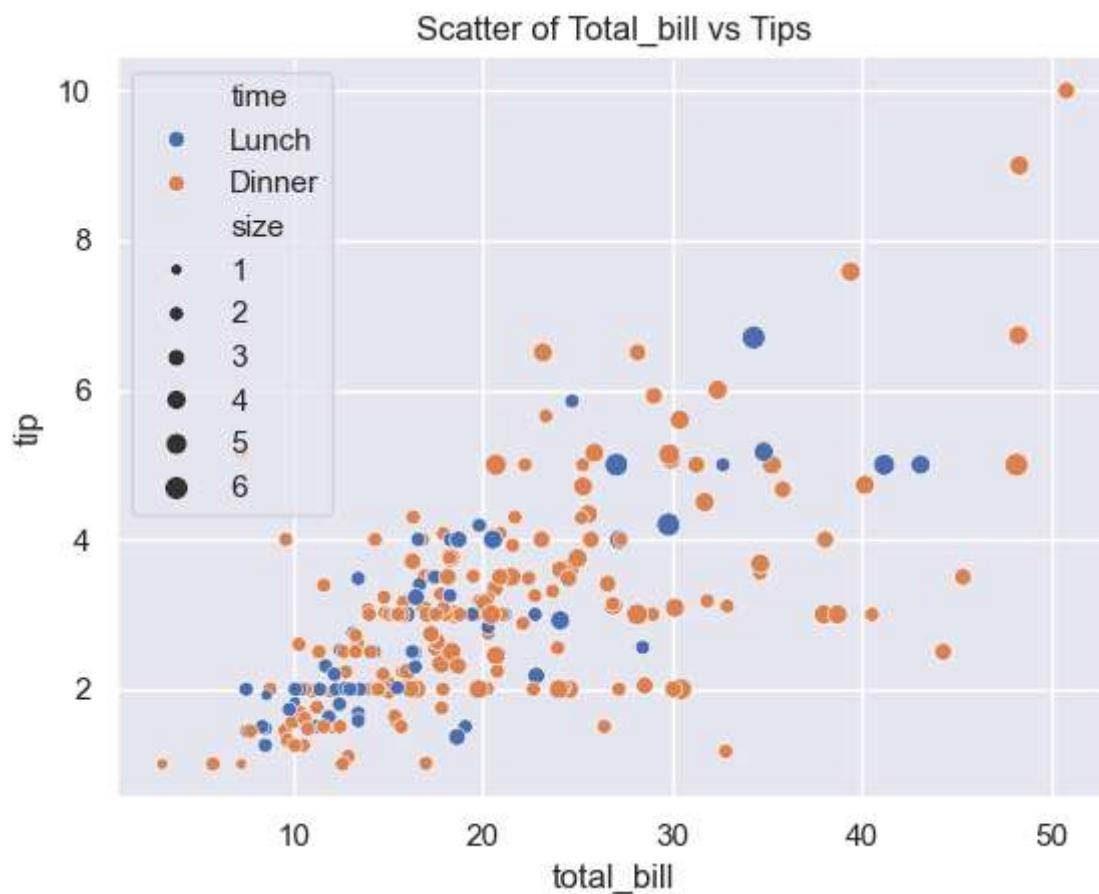
Out[10]: `b'C:\\Users\\suras'`

In [11]: `import matplotlib.pyplot as plt`
`plt.figure(figsize=(8,6))`

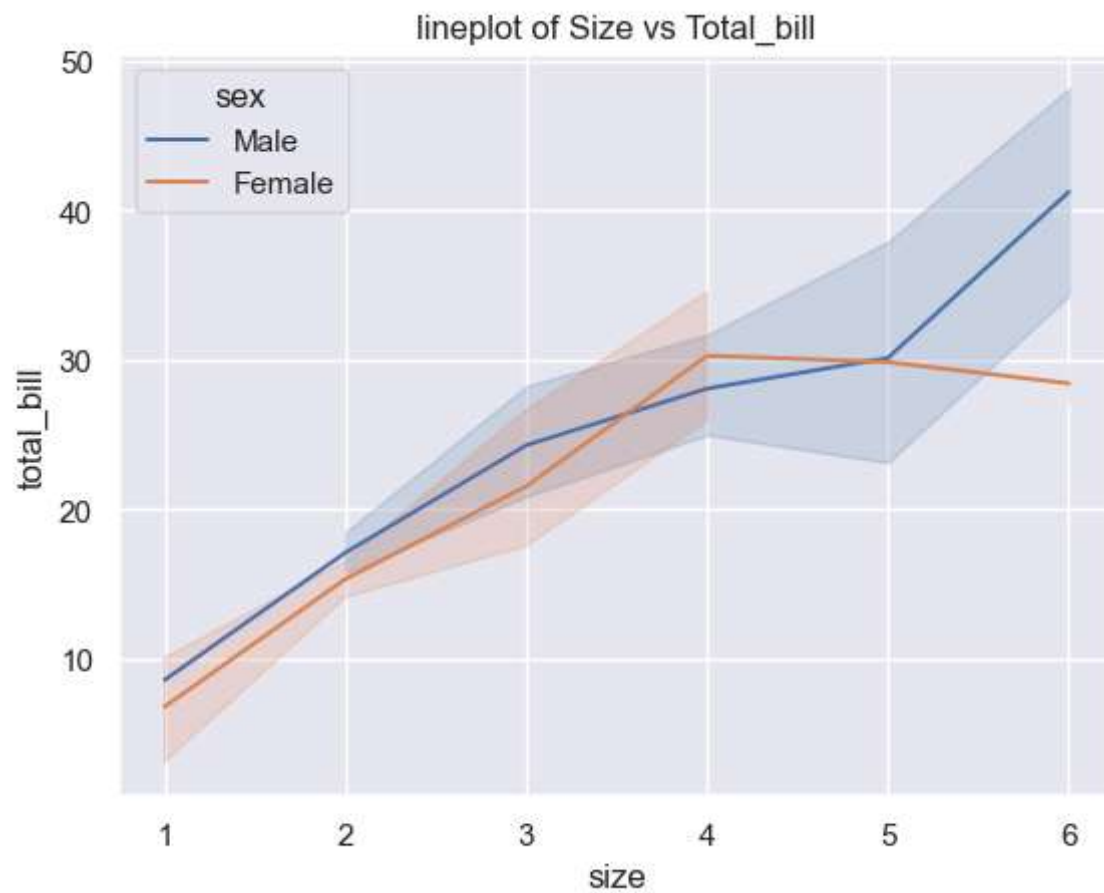
Out[11]: `<Figure size 800x600 with 0 Axes>`

<Figure size 800x600 with 0 Axes>

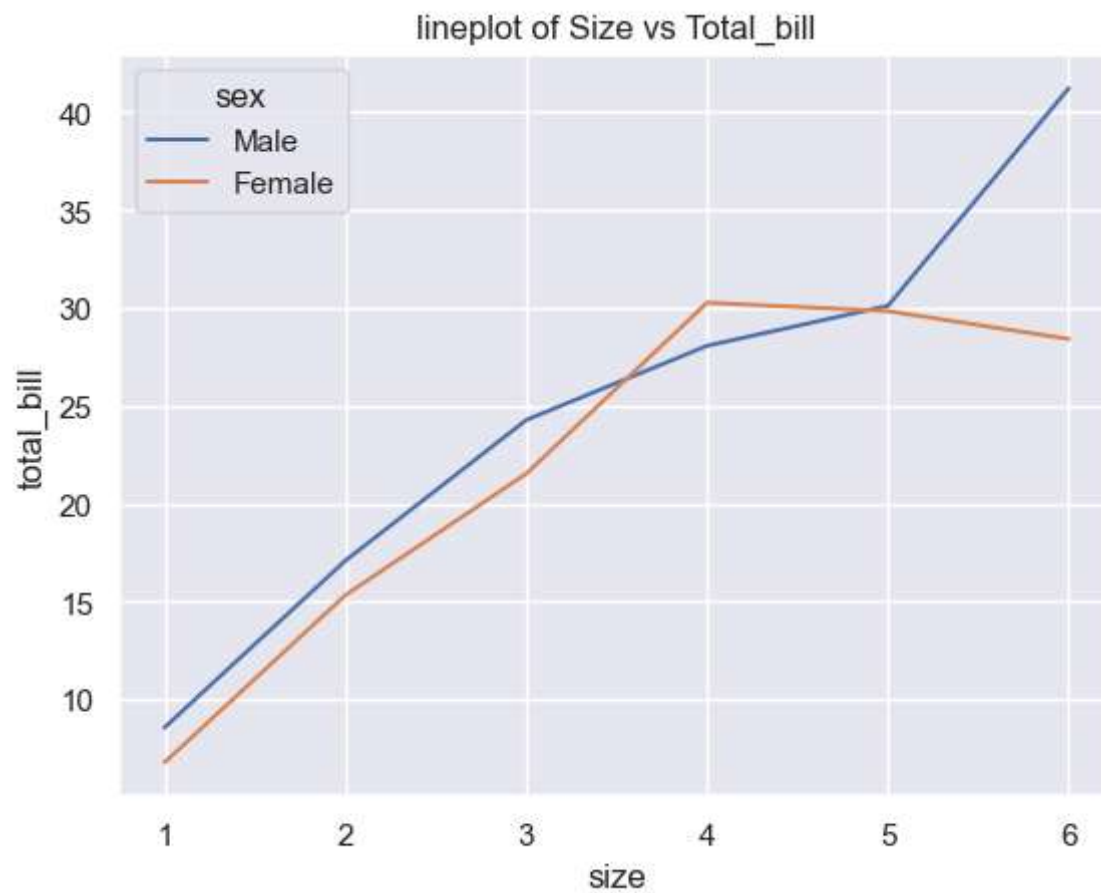
```
In [12]: sns.scatterplot(data=tips, x="total_bill", y="tip", size="size",  
hue="time")  
plt.title("Scatter of Total_bill vs Tips")  
plt.show()
```



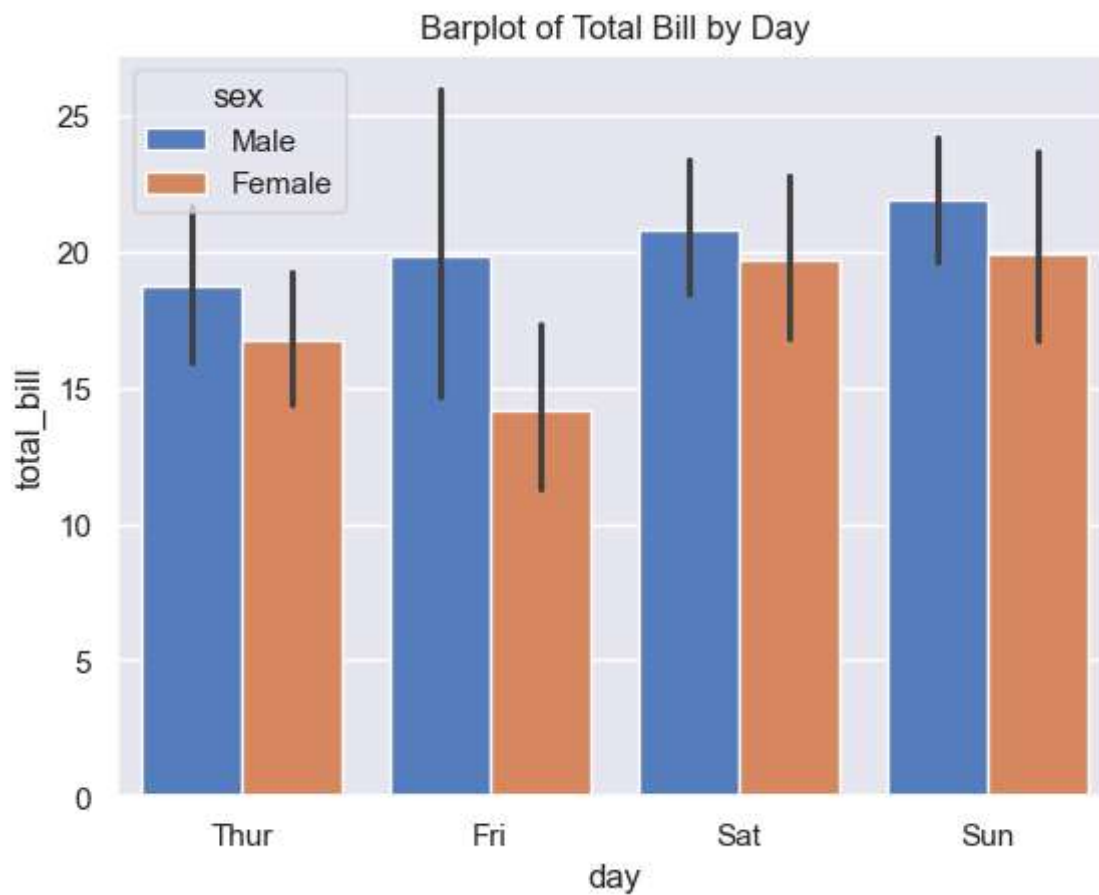
```
In [13]: sns.lineplot(data=tips, x="size", y="total_bill", hue="sex",  
markers='o')  
plt.title("lineplot of Size vs Total_bill")  
plt.show()
```



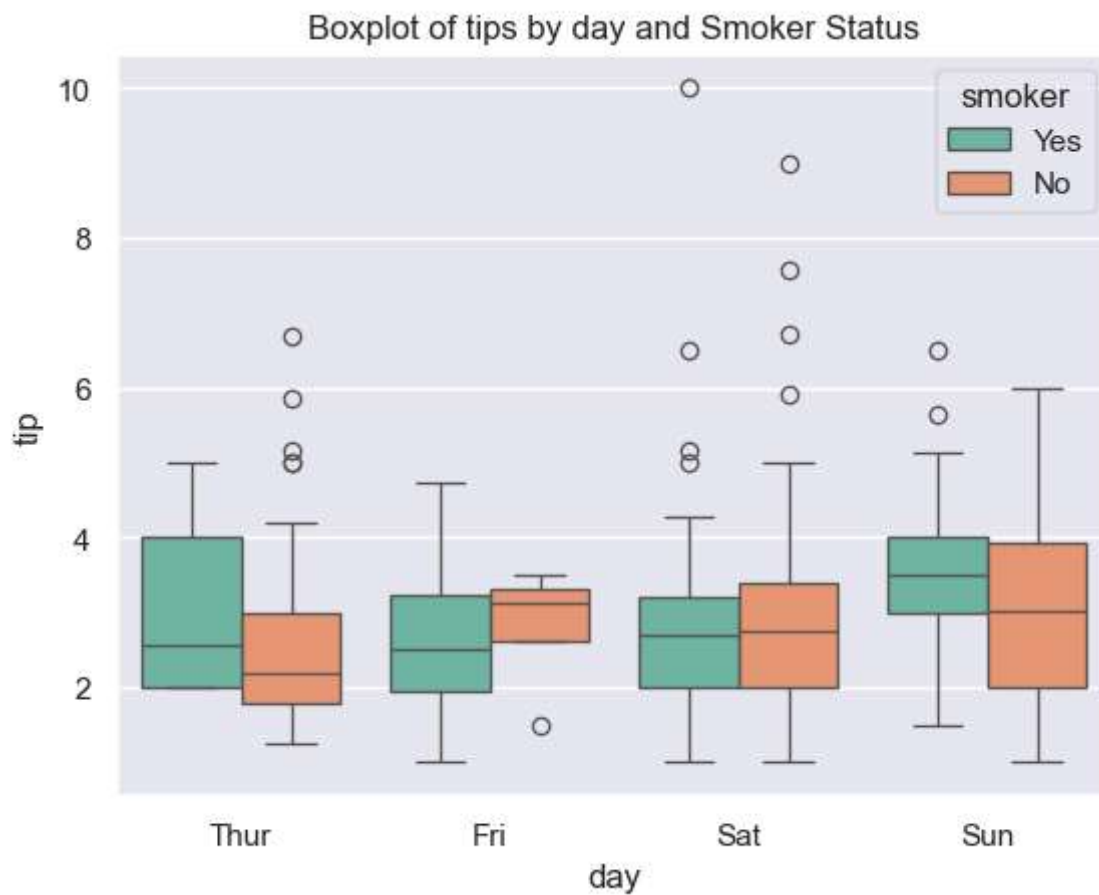
```
In [15]: sns.lineplot(data=tips, x="size", y="total_bill", hue="sex", ci=None,
markers='o')
plt.title("lineplot of Size vs Total_bill")
plt.show()
```



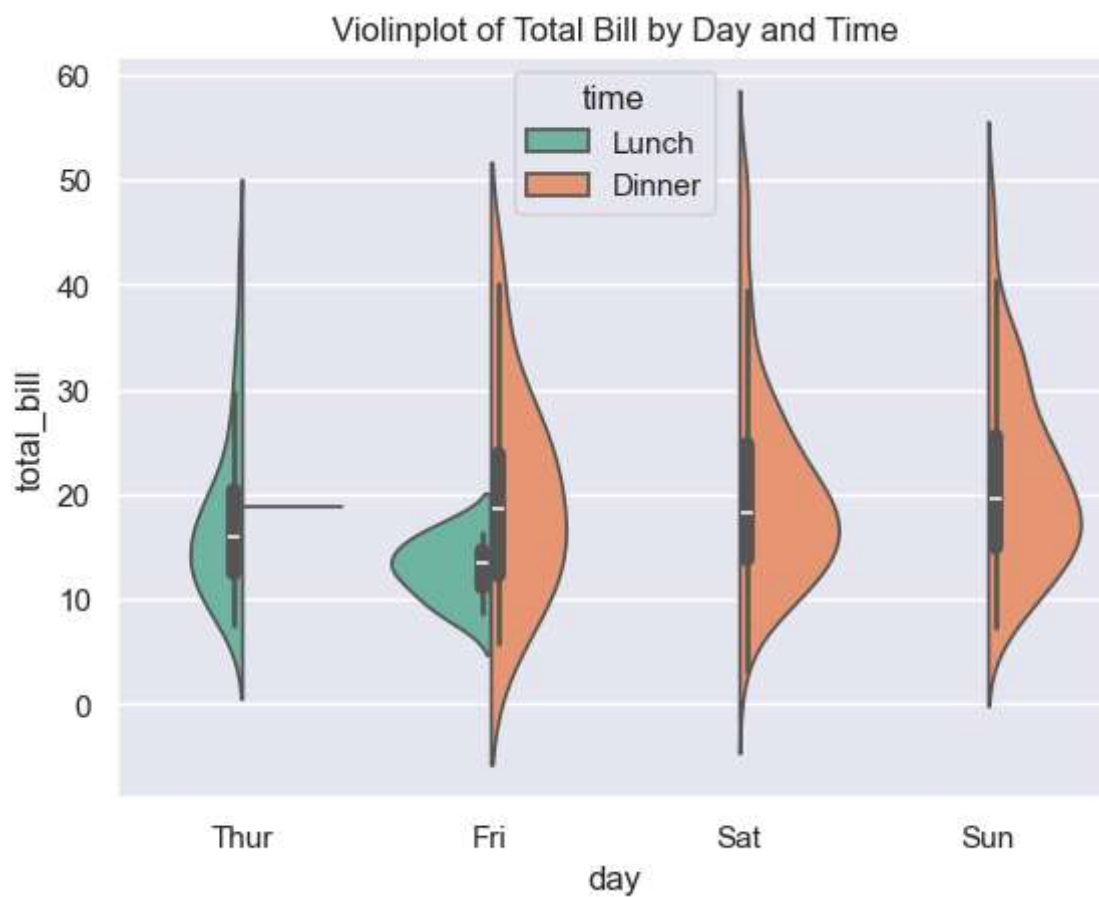
```
In [19]: sns.barplot(data=tips, x="day", y="total_bill", hue="sex",  
palette="muted")  
plt.title("Barplot of Total Bill by Day")  
plt.show()
```



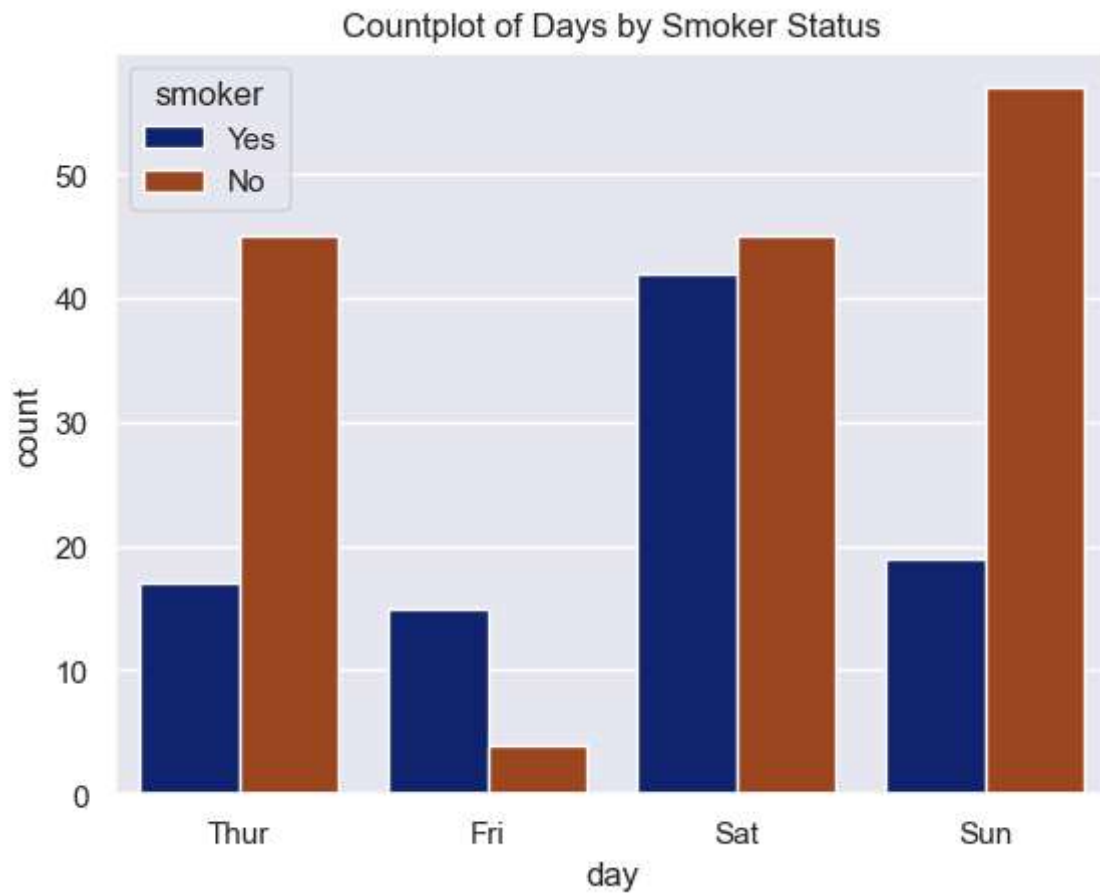
```
In [20]: sns.boxplot(data=tips, x="day", y="tip", hue="smoker", palette="Set2")  
plt.title("Boxplot of tips by day and Smoker Status")  
plt.show()
```



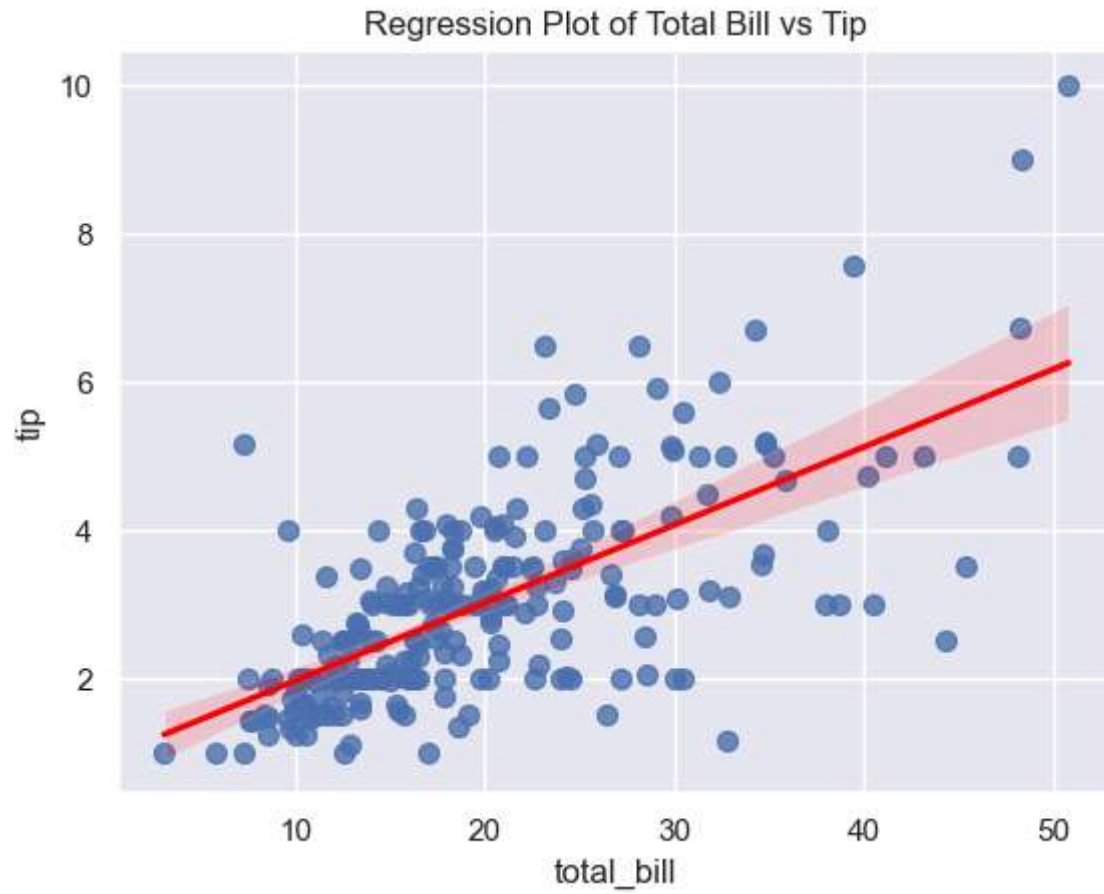
```
In [21]: sns.violinplot(data=tips, x="day", y="total_bill", hue="time",  
split=True, palette="Set2")  
plt.title("Violinplot of Total Bill by Day and Time")  
plt.show()
```



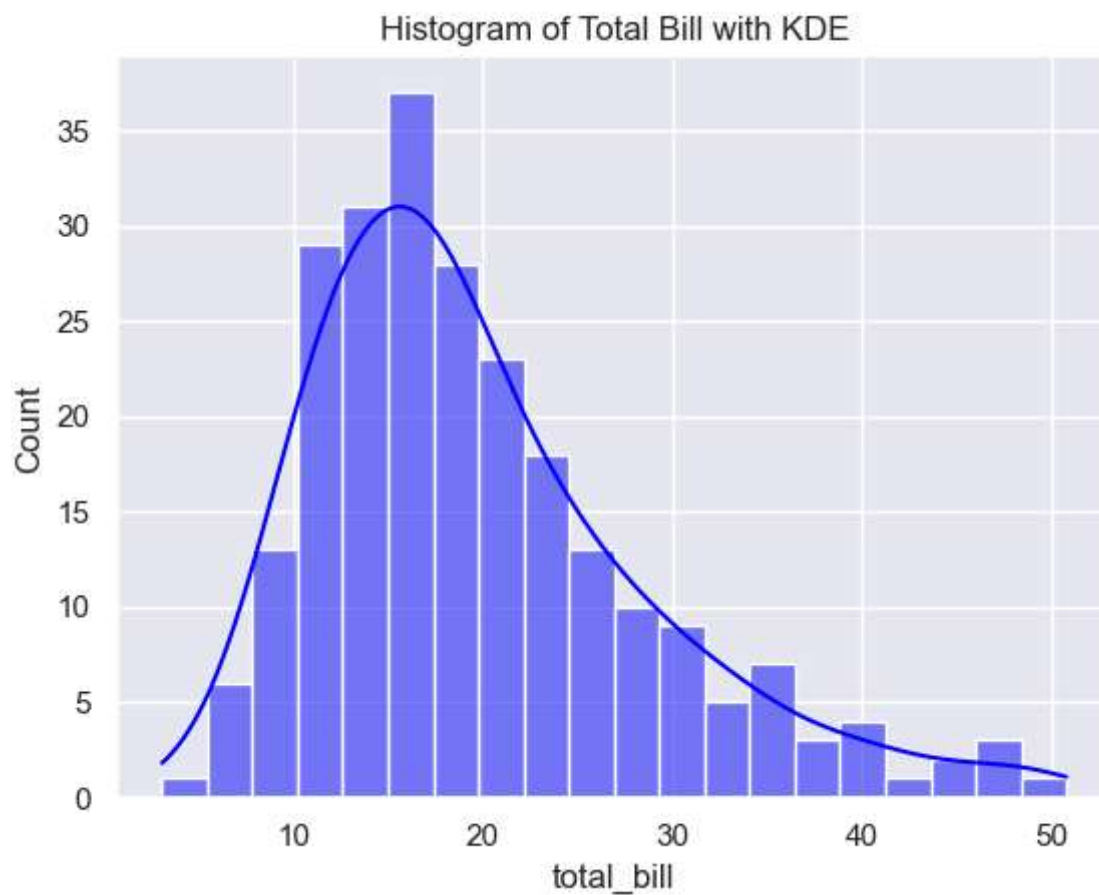
```
In [22]: sns.countplot(data=tips, x="day", hue="smoker", palette="dark")  
plt.title("Countplot of Days by Smoker Status")  
plt.show()
```

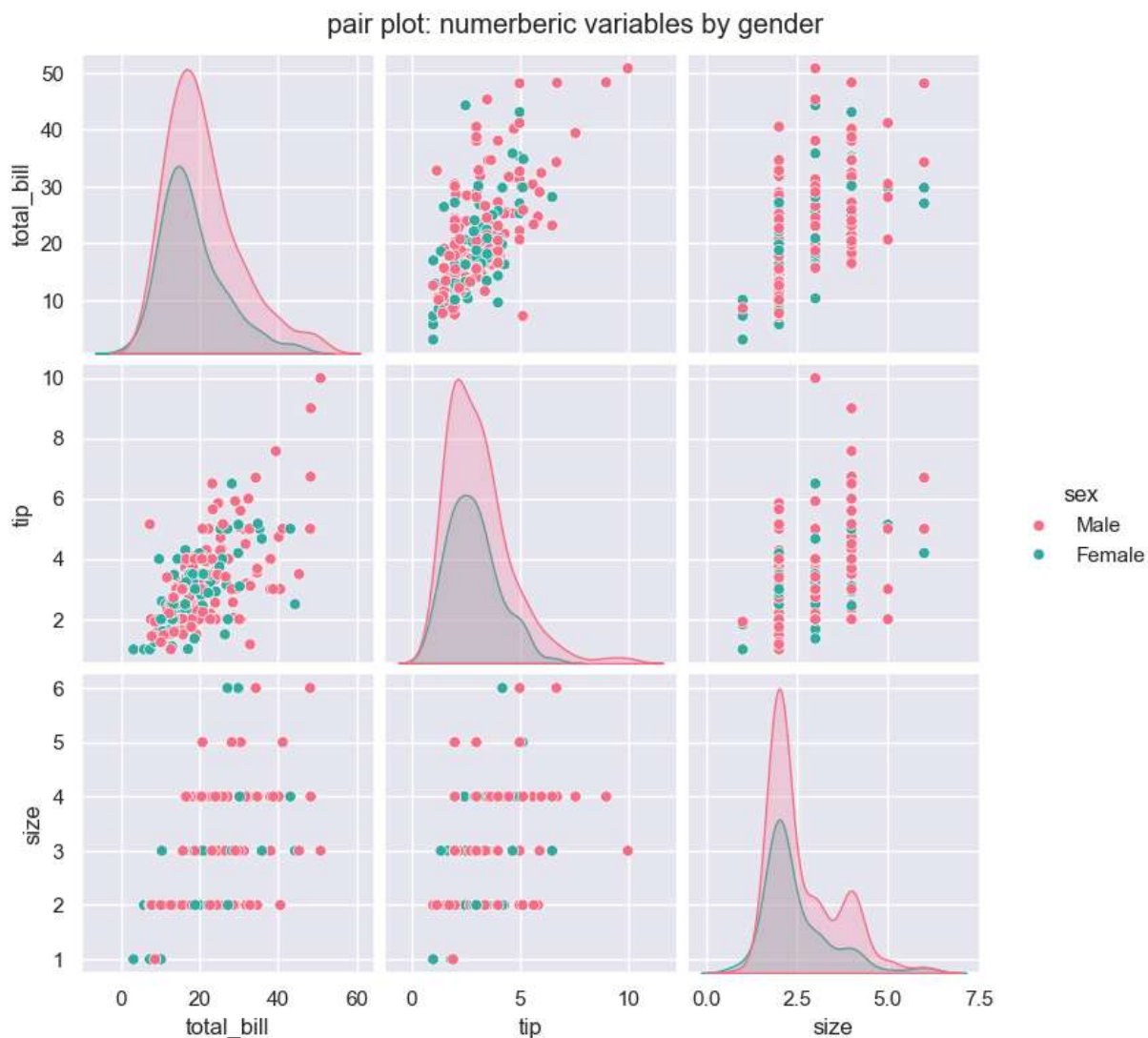
```
In [23]: sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50},  
line_kws={'color':'red'})  
plt.title("Regression Plot of Total Bill vs Tip")  
plt.show()
```



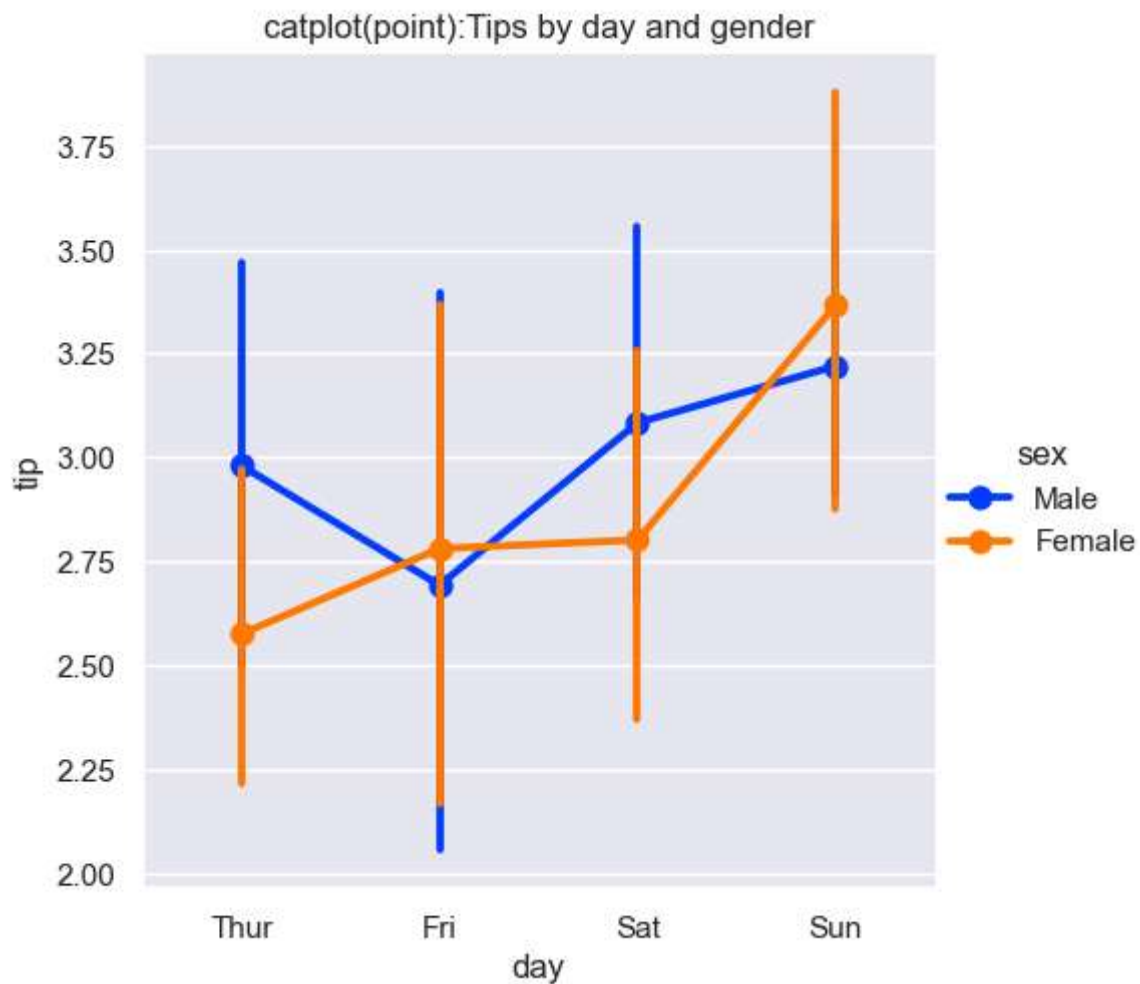
```
In [24]: sns.histplot(data=tips, x='total_bill', bins=20, kde=True,
color='blue')
plt.title("Histogram of Total Bill with KDE")
plt.show()
```



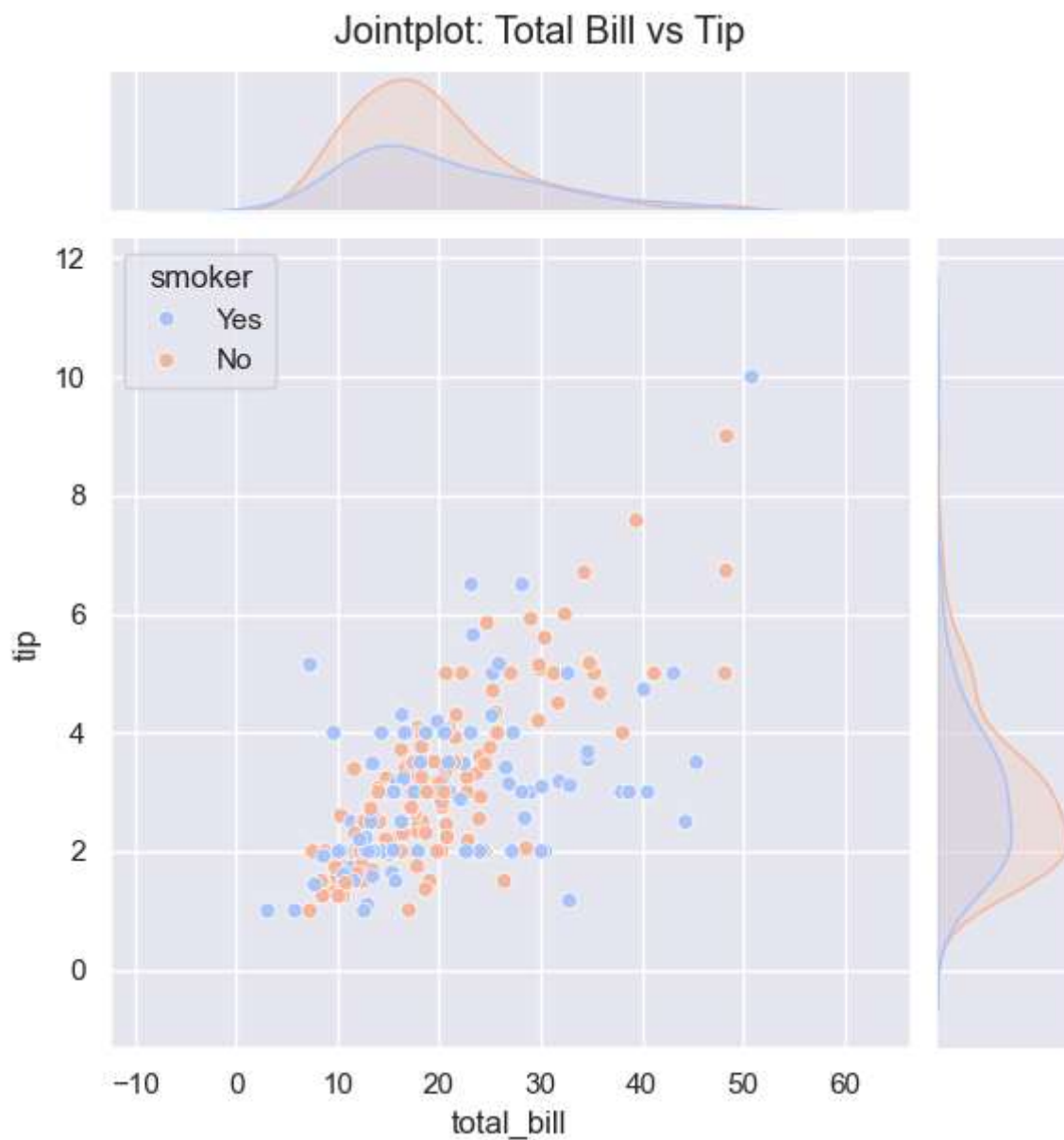
```
In [25]: sns.pairplot(tips, hue='sex', vars=["total_bill", "tip", "size"],  
palette='husl')  
plt.suptitle("pair plot: numerberic variables by gender", y=1.02)  
plt.show()
```



```
In [26]: sns.catplot(data=tips, x='day', y='tip', hue='sex', kind='point',
palette='bright')
plt.title("catplot(point):Tips by day and gender")
plt.show()
```

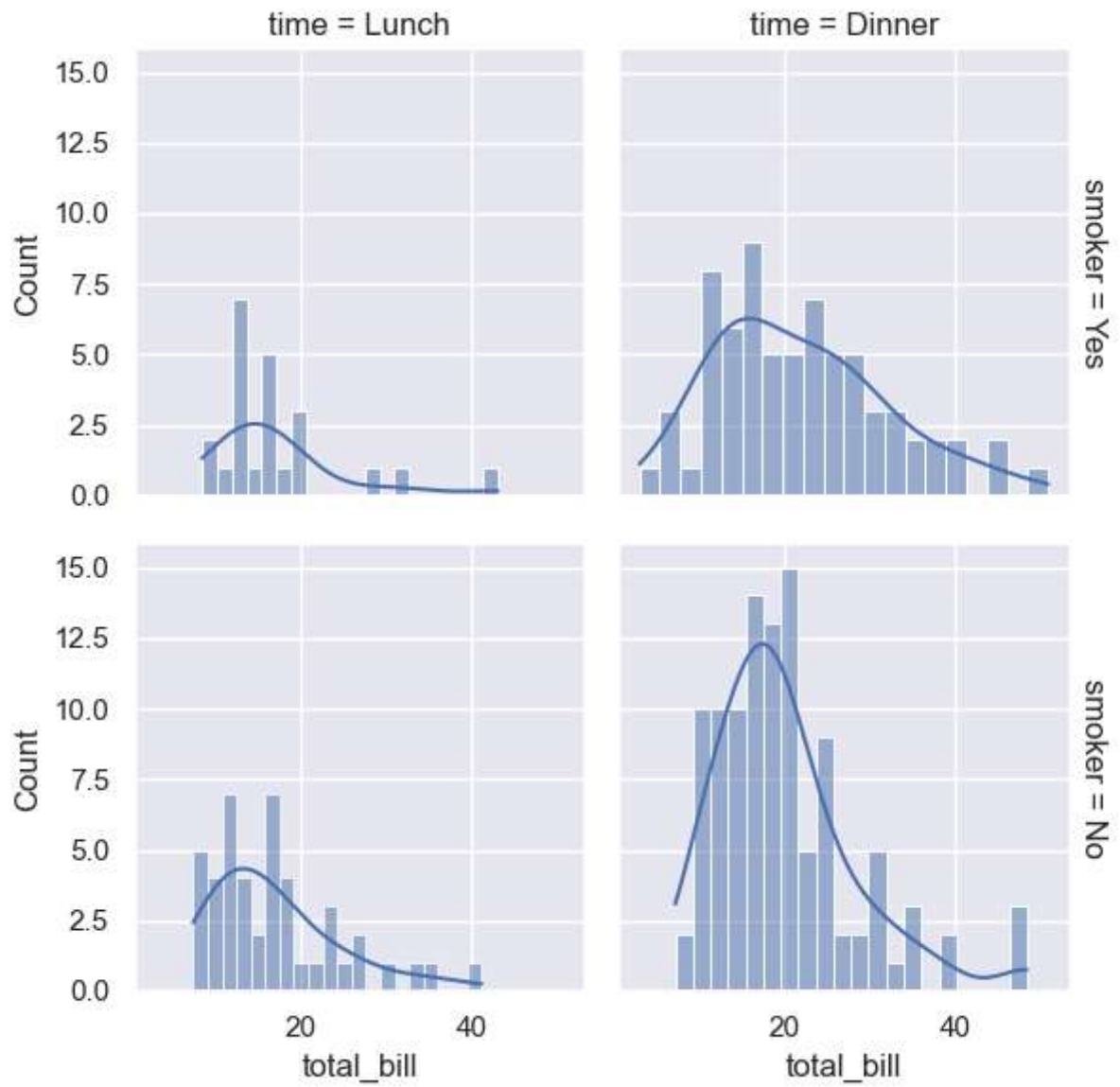


```
In [27]: sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter',  
hue='smoker', color='purple', palette='coolwarm')  
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)  
plt.show()
```



```
In [29]: g = sns.FacetGrid(tips, col='time', row='smoker',  
margin_titles=True).map(sns.histplot, 'total_bill', bins=20,  
kde=True).add_legend()  
g
```

```
Out[29]: <seaborn.axisgrid.FacetGrid at 0x153ce80afc0>
```



In []: